

SUMMARY

Parisa is a machine learning and computer vision researcher specializing in medical image analysis, generative modeling, multimodal learning, and foundation models for biomedical imaging. Her work includes clinically interpretable representation learning, with latent space embeddings demonstrating explainable tissue-level separability across cancer types. She has contributed to an independent external validation study of AI-driven medical image retrieval systems published in NEJM AI, proposing a framework of minimal requirements for clinical deployment. Beyond digital pathology, her broader research includes developing a novel VQ-GAN-based vector quantization framework for improved image reconstruction and feature representation, building end-to-end ML pipelines for large-scale biological imaging data, and applying deep learning to multiplex cancer imaging (CODEX) during her internship at Merck, where she fine-tuned U-Net-based models for cellular phenotyping in tumor microenvironments and developed M3-JEPA, a multimodal joint-embedding predictive framework aligning H&E histology with spatial transcriptomics for cross-modal retrieval. Her research has received over 290 citations. With strong expertise in PyTorch, distributed training, and reproducible pipeline development, she is well-positioned to contribute to translational AI research at the intersection of medical imaging AI and clinical deployment.

EXPERIENCE

- **Merck & Co., Inc.** West Point, PA
Image Data Analytics Intern *Summer 2026 - Present*
 - Developed a multimodal framework to predict spatial gene expression from H&E histology by aligning whole-slide images with paired spatial transcriptomics profiles, trained on the HEST-1k dataset of matched histology and spatial transcriptomic samples.
 - Adapted M3-JEPA, a joint-embedding predictive architecture, using frozen foundation-model encoders (H-optimus-0 for histology and Novae for spatial transcriptomics) with a mixture-of-experts predictor trained on cached embeddings via a combined regression and contrastive objective in PyTorch.
 - Established an end-to-end training and evaluation pipeline for cross-modal H&E-to-gene-expression retrieval and downstream tissue classification, benchmarking the learned embeddings to analyze morphology-expression alignment.
- **Merck & Co., Inc.** South San Francisco, CA
Computer Vision and AI Research Intern *Summer 2025*
 - Conducted in-depth analysis of large-scale, high-resolution medical (CODEX) imaging datasets of cancer tissue, identifying significant class imbalance across cell types (tumor and immune cells) that adversely affected model performance.
 - Designed and applied a domain-specific data augmentation pipeline and employed transfer learning by fine-tuning a pretrained U-Net architecture in PyTorch, adapting it to domain-specific multiplex cancer imaging data following cell-level segmentation using DeepCell, to improve classification performance of cellular phenotypes in imbalanced settings.
 - Compared the classification accuracy of the fine-tuned model against a state-of-the-art self-supervised foundation model for spatial proteomics (KRONOS, Harvard Medical School), demonstrating superior performance on multiplex cancer imaging data.
- **University of Texas at Arlington** Arlington, TX
Graduate Research Assistant *Aug 2022 - Present*
 - Developed Dual Codebook VQ, a novel vector quantization framework for learning compact and expressive image features, leveraging a transformer to optimize a global codebook and achieving strong performance with 50% fewer code vectors than prior CLIP-based approaches. Integrated the method into a VQGAN architecture with a ResNet encoder and trained on large-scale datasets (MS-COCO, ADE20K, CelebA-HQ). Achieved up to 57% improvement in reconstruction quality (FID), while significantly improving feature extraction efficiency and representation quality for downstream image analysis. Built fully reproducible training pipelines using PyTorch Lightning.
 - Conducted the first independent external validation of four state-of-the-art histopathology slide search engines on real patient cases from a clinical health system, applying a structured evaluation framework across multiple clinically defined retrieval tasks on unseen datasets, and contributed to a proposed framework of minimal requirements for clinical deployment of WSI search systems published in NEJM AI.

- Developed a VAE-based pipeline for compression and reconstruction of H&E whole slide images (WSIs) across four cancer types (brain, breast, lung, colon) using the NCI GDC dataset, achieving a state-of-the-art 1:512 compression ratio while preserving downstream clinical classification performance, and demonstrated clinically interpretable latent space embeddings via UMAP that successfully separate distinct cancer tissue types — with potential applications in rare cancer subtyping and hospital-scale clinical image search.
- Developed an end-to-end pipeline for longitudinal modeling of temporal dynamics in live-cell biological imaging data of *Drosophila*, including preprocessing (denoising), cell segmentation using Cellpose 2.0, feature extraction, and regression modeling with XGBoost. Extracted quantitative cell-level features to capture spatiotemporal patterns and enable predictive modeling of dynamic biological processes. The pipeline was designed as a robust and reusable workflow for consistent analysis across large-scale imaging datasets.

EDUCATION

- **University of Texas at Arlington (UTA)** Arlington, TX
Ph.D. in Computer Science 2022 – 2026
- **Iran University of Science and Technology (IUST)** Tehran, Iran
Master of Electrical Engineering 2018–2021
- **Babol Noshirvani University of Technology (NIT)** Babol, Iran
Bachelor of Electrical Engineering 2012–2017

SKILLS AND ACTIVITIES

- **Programming Languages:** Python, MATLAB, Git, familiar with C++ and SQL
- **Python Packages:** NumPy, Pandas, Matplotlib, Scikit-learn, SciPy, Seaborn, OpenCV
- **Deep Learning Frameworks:** PyTorch (PyTorch Lightning), Keras, TensorFlow, Distributed Training (multi-GPU), Transfer Learning & Fine-Tuning
- **Software and IDEs:** Jupyter, Google Colab, PyCharm
- **Generative and Multimodal AI:** Autoencoders (e.g. VAE, VQ-VAE, VQ-GAN), Conditional Generative Models, M3-JEPA, DINO, Diffusion models, Multimodal Transformers (e.g. CLIP), Vision Transformers, U-Net
- **Computer Vision Applications:** Classification, Semantic Segmentation, Object Detection
- **Medical Imaging & Domain Expertise:** Digital Pathology, Spatial Transcriptomics, Medical Image Analysis (H&E, multiplex CODEX), QuPath, Biomarker Discovery, Single-Cell Phenotyping, Computational Biology, Computational Genomics
- **Data Handling:** Data Cleaning, Visualization, Augmentation
- **Language Proficiency:** English (Fluent), Farsi (Native)

HIGHLIGHTED PUBLICATIONS

(* denotes co-first authorship)

1. **Parisa Boodaghi Malidarreh**, Jillur Rahman Saurav, Thuong Le Hoai Pham, Amir Hajighasemi, Anahita Samadi, Saurabh Shrinivas Maydeo, Mohammad Sadegh Nasr, Jacob M. Luber. Dual Codebook VQ: Enhanced Image Reconstruction with Reduced Codebook Size. *Prepared for Submission*, arXiv:2503.10832, 2025.
2. **Parisa Boodaghi Malidarreh**, Priyanshi Borad, Biraaj Rout, Anna Makridou, Shiva Abbasi, Mohammad Sadegh Nasr, Jillur Rahman Saurav, Kelli D Fenelon, Jai Prakash Veerla, Jacob M. Luber, et al. Machine Learning-Driven Prediction of Spatiotemporal Dynamics of Active Nuclei During *Drosophila* Embryogenesis. *International Journal of Molecular Sciences*, 26(21):10338, 2025.
3. Helen H Shang, Mohammad Sadegh Nasr, Jai Prakash Veerla, Jillur Rahman Saurav, Amir Hajighasemi, **Parisa Boodaghi Malidarreh**, Manfred Huber, Chace Moleta, Jitin Makker, Jacob M. Luber. Histopathology Slide Indexing and Search—Are We There Yet? *NEJM AI*, 1(5):AIcs2300019, 2024.
4. Kelli D Fenelon*, Priyanshi Borad*, Biraaj Rout*, **Parisa Boodaghi Malidarreh***, Mohammad Sadegh Nasr, Jacob M. Luber, Theodora Koromila. Su(H) Modulates Enhancer Transcriptional Bursting in Prelude to Gastrulation. *Cells*, 13(21):1759, 2024.

SOCIAL MEDIA

- **Personal Website:** <https://parisa-boodaghi.github.io>
- **LinkedIn:** <https://www.linkedin.com/in/Parisa-Boodaghi>
- **Google Scholar:** <https://www.scholar.google.com/in/Parisa-Boodaghi>